

Research on obstacle avoidance of multi-AUV cluster formation based on virtual structure and artificial potential field method[★]

Wenjun Ding^{a,b,*}, Lei Zhang^a, Guozong Zhang^a, Chiyu Wang^a, Yajun Chai^a,
Tihao Yang^c, Zhaoyong Mao^a

^a Unmanned System Institute, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, PR China

^b Research Institute of Xi'an Jiaotong University, Hangzhou, Zhejiang 311200, PR China

^c School of Aeronautics of Northwestern Polytechnical University, Xi'an, Shaanxi 710072, PR China

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ABSTRACT

The formation of autonomous underwater vehicle (AUV) in the three-dimensional environment is very important for carrying out complex ocean tasks. During formation sailing, it is necessary to consider avoiding obstacles while maintaining the formation. In this paper, an AUV formation obstacle avoidance method based on virtual structure and artificial potential field method is proposed. Firstly, a complete mathematical modeling of AUV is carried out, and a sliding mode position tracker based on virtual position points is designed based on this model. Then the tracking points of each AUV in the formation are calculated according to the expected trajectory and formation feature structure to ensure that the spacing between AUVs meets the requirements of the formation. Finally, the artificial potential field method is introduced into the position tracker to avoid obstacles between the obstacle and the AUV. Five groups of simulation experiments of parallel formation, vertical formation, cross formation, triangular formation and unknown complex environment are designed. The formation scalability of the algorithm is simulated and compared with the virtual leader method in related references. Finally, KKSwarm platform is used to conduct a real simulation in 2D environment to verify the effectiveness of the proposed algorithm. The experimental results show that the proposed method can effectively realize the formation navigation of multiple underwater vehicles and successfully avoid obstacles, which provides an effective method to solve the problem of the formation obstacle avoidance of underactuated AUV.

1. Introduction

Autonomous Underwater Vehicle (AUV) is a kind of unmanned underwater vehicle capable of carrying out underwater tasks autonomously. AUV plays an important role in scientific research, resource utilization, Marine safety, transportation, military application, education and science popularization and other fields, and is of great significance for human understanding, protection, and utilization of Marine resources [12].

However, the resources and search range of a single AUV are very limited, and often cannot meet the requirements of diverse search tasks, which often require multiple AUVs to work together to complete the task. The introduction of multiple AUV clusters not only can

[★] Correspondence to.

E-mail address: dingwenjun@nwpu.edu.cn (W. Ding).

greatly improve the search efficiency, but also can reduce the task burden of each AUV. Therefore, the transition from single AUV search to AUV cluster search is a sure way to achieve more efficient and intelligent underwater mission detection.

Underwater unmanned clusters, consisting of multiple underwater unmanned vehicles (AUVs), are a new type of underwater detection technology that achieves specific tasks through autonomous group collaboration. Compared with a single AUV system, AUV clusters have the following advantages [18]:

1. Diversified system spatial distribution: multiple AUVs can be deployed in a reasonable spatial formation for different waters to carry out the mission.
2. Low cost: compared with a single powerful underwater vehicle, the cost of multiple underwater cluster submersibles with simple functions is lower.
3. High system fault tolerance: mechanical failure of a single submersible can directly lead to mission failure. If the AUVs fail to complete the mission, the cluster system can continue to complete the mission through formation adjustment.

With the continuous improvement of computer technology and performance, the cluster method of underwater vehicles has been developed rapidly under the joint research of global experts and scholars. As an important aspect of multi-AUV system, formation problem has attracted the attention of many scholars. At present, many formation algorithms are used, including Leader-follower method [7], virtual structure-based method [23], artificial potential field-based method [24] and behavior-based formation method [2].

The Leader-follower method was first proposed by Cruz [5], the formation members are divided into a leader and a follower, the leader runs according to the trajectory set by the mission, and the follower adjusts its position according to the distance and orientation relative to the leader to maintain the current formation. However, the disadvantage is that the leader bears more computational pressure, and if the leader's trajectory is shifted, it will cause the overall trajectory shift. A formation control method of AUV group combining consensus theory and leader-follower method under communication delay is proposed [9]. The method of virtual leader is adopted to reduce the computational pressure of the leader and considers the effects of underwater communication delays [20]. Reference [21] mainly adopts the formation algorithm of virtual leader, and studies the problem of multi-AUV cost-preserving formation tracking control based on switching topology communication, and realizes the formation tracking control of multi-AUV system.

The Virtual structure based method was first proposed by [17], the kinematic analysis is performed according to the specified virtual geometry, and the corresponding motion characteristics of the virtual key point on the shape are deduced, and the algorithm is used to realize the AUV tracking the key point and calculating its error with the virtual key point in real time and adjusting it to achieve formation control [23]. The advantage of this method is that there is formation feedback among robots to maintain the formation, which reduces the computational burden of the central robot, and each robot takes the corresponding computational load. However, the disadvantage is that the algorithm lacks flexibility because of the need for formation maintenance and poor obstacle avoidance performance, which is usually combined with other obstacle avoidance algorithms to complete formation control in complex underwater environments [6]. At present, it is mostly applied to aerial robot formation, and less applied in AUV formation [11]. A method of finite-time position and attitude tracking control method combined with artificial potential field and virtual structure is presented by Qiang [16]. The results indicate that the multi-AUV system can form a formation without collision from random initial positions. But it only studied formation control in two dimensions. Reference [22] proposes a swarm control algorithm for underwater vehicles based on maneuverability prediction. A virtual structure is constructed to organize the formation, and the desired trajectory of the virtual structure is tracked by the attraction potential function, so as to solve the problem of insufficient organizing ability of the artificial potential field.

Artificial potential field based method: first proposed by Khatib [10], by constructing an artificial force field in a known environment, the target point is attractive to the robot and the obstacle is repulsive to the robot, the robot is set as a potential function, and the formation control is achieved by minimizing the individual potential field [8]. The advantage of this method is that the obstacles can be effectively avoided, but the disadvantage is that it is difficult to find a suitable potential function. At present, it is mostly used in combination with other formation control algorithms to achieve formation obstacle avoidance. A novel kind of AUV path planning algorithm is proposed by combining the velocity synthesis algorithm and artificial potential field method. The improved artificial potential field function is used to avoid obstacles, and the velocity synthesis algorithm is used to achieve the optimal path [4]. Noguchi and Maki [13] proposed a combination of artificial potential field method and reinforcement learning for path planning. Reference [14] studies the problem of AUV formation obstacle avoidance under the interference of ocean currents. A variable formation reconstruction and obstacle avoidance control scheme based on affine transformation and improved artificial potential field is designed, so that AUV can avoid both static and dynamic obstacles and maintain the desired time-varying formation under external interference.

Behavior based formation method: first proposed by Brooks [1], the control idea of this method is: decompose the formation task into: converging to the target point, obstacle avoidance, formation control and other basic behaviors, and achieve AUV formation control through behavior fusion. The advantage of this method is that it is easy to realize distributed control and the system is more resilient. The disadvantage is that the overall behavior of the formation system cannot be clearly defined, which is not conducive to the stability analysis of the system [19]. Chen [3] proposes a path planning method based on behavioral decision-making. This algorithm can solve the path with the lowest energy loss in the achievable path.

Since this paper involves AUV formations, the performance analysis of common AUV formations is carried out. The common formations are shown in the following

Parallel formation is the simplest form of AUV cluster formation, in which all AUVs are arranged on the same line with equal

spacing. The advantage is simple and easy to achieve, the disadvantage is that collision is easy, so the formation safety is low, and the obstacle avoidance ability is poor in the complex environment. From the perspective of energy loss, because the spacing between AUVs is small and located in the same straight line, the water flow interference between each other is large, so the energy consumption is more. In addition, because the AUV is concentrated in a straight line, the exploration range is relatively wide, and considering its high energy consumption, this formation is suitable for small-range carpet exploration tasks.

Vertical formation means that AUVs are arranged in a straight line. This formation is easy to deploy and ensures communication and coordination between AUVs. However, the exploration range of the linear formation is very limited, and only the area on the linear path can be detected, so it takes a longer time to detect the entire sea area, and its detection performance does not bring advantages because of its cluster formation, so it is not suitable for any exploration mission. In addition, due to the small spacing between AUVs, collisions are prone to occur, and the safety of the formation is lower. However, in terms of avoiding obstacles on narrow roads, the vertical formation has obvious advantages, and track tracking can be done along narrow straight or curved roads. Therefore, the linear array is mainly used for avoiding obstacles on narrow roads.

Cross formation is composed of two straight lines, and there are a lot of cross paths between the AUVs of the cross formation, which leads to the frequent acceleration and deceleration of some AUVs to avoid collision, thus increasing the energy consumption. Moreover, because of the large cluster density of AUV, the computational complexity of the obstacle avoidance algorithm is easily increased, resulting in the decrease of obstacle avoidance efficiency. However, the detection range of horizontal cross array formation is wide, and there are repeated exploration paths between the middle AUV and the tail AUV, which does not fully play the exploration advantages brought by cluster exploration. However, in the research direction of robot encircle in underwater attack and defense, the cross array can control the motion of the intermediate AUV on the horizontal plane well, so it is suitable for underwater robot encircle on the two-dimensional plane.

Triangular formation is the combination of multiple AUV tree arrangements into triangles. The shape formed by the outer AUVs mimics the V-shaped formation based on the flight principle of wild geese. According to the relevant knowledge of fluid mechanics, the streamlined formation can minimize the flow resistance of the AUVs at the back of the team, thus reducing energy consumption. Moreover, the distance between AUVs in the triangular formation is larger. The advantage of low formation density makes its formation safety and detection performance better than that of cross array. However, in the aspect of formation obstacle avoidance, the steering ability of triangular formation is insufficient and the obstacle avoidance efficiency is low. Therefore, the triangular formation is suitable for large-scale exploration missions at long distances.

Through the analysis of the above common formations, it can be found that different formations have different performance in different evaluation indicators, so in practical application, it is necessary to select the best formation according to the specific task requirements, and therefore, there are studies on the direction of AUV cluster formation transformation in different situations.

Based on the above four basic formations, a cluster formation method based on the combination of virtual structure and artificial potential field is proposed in this paper. The tracking structure points of AUV are solved according to the expected trajectory and virtual structure method to ensure that the spacing between AUVs meets the formation requirements. A 3D sliding mode trajectory tracking controller is designed to realize the AUV tracking of the corresponding structural points. At the same time, the method combined with artificial potential field can guide AUV to avoid obstacles during formation sailing. By combining these methods, the multi-AUV cluster formation navigation is realized.

In this paper, the second chapter introduces how to establish and simplify the 5-DOF mathematical model of AUV. In chapter 3, a 3D trajectory tracking controller for underactuated AUV is designed based on sliding mode control. In chapter 4, the obstacle avoidance controller is designed based on the artificial potential field method. In chapter 5, cluster controller based on virtual structure method is designed. In Chapter 6, simulation experiments are designed to verify the effectiveness of the algorithm, and finally summarizes the paper.

2. Underactuated AUV 5-DOF mathematical model

The attitude of the AUV is described by defining the fixed coordinate system $\{O_A\}$ and the object coordinate system $\{O_B\}$. The relevant motion parameters are defined as shown in Fig. 2.

Parameters η and v are used to describe all motion parameters in the fixed coordinate system $\{O_A\}$ and the object coordinate system $\{O_B\}$ respectively. For details, see Eq. (1).

$$\begin{cases} \eta = [\eta_1, \eta_2]^T \\ \eta_1 = [x, y, z]^T \\ \eta_2 = [\varphi, \theta, \psi]^T \end{cases} \quad \begin{cases} v = [v_1, v_2]^T \\ v_1 = [u, v, w]^T \\ v_2 = [p, q, r]^T \end{cases} \quad (1)$$

AUV is regarded as a rigid body, and its attitude in three-dimensional space mainly refers to the rotation relationship between the body coordinate system and the fixed coordinate system of the earth. This rotation relationship can be described in terms of Euler angles. The fixed coordinate system can be coexisting with the object coordinate system by rotating the axis at a certain Angle. In this way, the correlation between the motion attitude of the fixed coordinate system and the object coordinate system can be established. The detailed conversion process is shown in Fig. 3.

Therefore, the transformation relationship between the position vector and the linear velocity vector of the underwater vehicle can be obtained, as shown in Eq. (2).

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = J_1(\eta_2) = R_{z_A}(\psi)R_{y_A}(\theta)R_{x_A}(\varphi) \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad (2)$$

By following the rotation order in Fig. 3, we can get the relationship between the angular velocity component v_2 and the attitude Angle change rate $\dot{\eta}_2$ in the object coordinate system.

$$v_2 = \dot{\psi} \cdot {}^B\vec{z}_B + \dot{\varphi} \cdot {}^B\vec{y}_B + \dot{\theta} \cdot {}^B\vec{x}_B \quad (3)$$

Where:

$$\begin{cases} {}^B\vec{x}_B = [1, 0, 0]^T \\ {}^B\vec{y}_B = {}^B_R(\varphi)^{-1} \cdot [0, 1, 0]^T \\ {}^B\vec{z}_B = [{}^B_R(\theta) \cdot {}^B_R(\varphi)]^{-1} \cdot [0, 0, 1]^T \end{cases} \quad (4)$$

Therefore, the conversion relationship between the angular velocity component and the attitude angle conversion rate can be obtained by the following equation.

$$\begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\varphi\tan\theta & \cos\varphi\tan\theta \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi/\cos\theta & \cos\varphi/\cos\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = J_2(\eta_2) \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (5)$$

In summary, the six-degree-of-freedom kinematic equation of underactuated AUV is as follows:

$$\begin{bmatrix} \dot{\eta}_1 \\ \dot{\eta}_2 \end{bmatrix} = \begin{bmatrix} J_1(\eta_2) & 0_{3 \times 3} \\ 0_{3 \times 3} & J_2(\eta_2) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \Leftrightarrow \dot{\eta} = J(\eta)v \quad (6)$$

The 6-degree-of-freedom dynamic equation of the commonly used underwater vehicle in the object coordinate system is as follows:

$$M\dot{v} + C(v)v + D(v)v + g(\eta) = \tau + \tau_d \quad (7)$$

Where: M is the AUV mass matrix, $C(v)$ is the Koch force matrix, $D(v)$ is the damping parameter matrix; $g(\eta)$ is the recovery force matrix; $\tau = [\tau_u, 0, 0, 0, \tau_q, \tau_r]$ is the AUV drive configuration, $\tau_d = [\tau_{d1}, \tau_{d2}, \tau_{d3}, \tau_{d4}, \tau_{d5}, \tau_{d6}]$ is the model uncertainty and environmental disturbance. In order to simplify the multi-free mathematical model of AUV, some assumptions are made in this paper due to the high complexity and coupling of the dynamic equations of AUV in three-dimensional space. I. Treating AUVs as rigid bodies. II. Assuming that the main body of an AUV is basically left and right symmetric, its shape and mass distribution are the same in all directions. III. The roll freedom of the AUV during navigation is ignored. Therefore, the kinematic model of AUV can be derived and simplified using the above assumptions, and its differential expression is shown in Eq. (8).

$$\begin{cases} \dot{x} = u\cos\psi\cos\theta - v\sin\psi + w\cos\psi\sin\theta \\ \dot{y} = u\sin\psi\cos\theta + v\cos\psi + w\sin\psi\sin\theta \\ \dot{z} = -u\sin\theta + w\cos\theta \\ \dot{\theta} = q \\ \dot{\psi} = r/\cos\theta \\ \dot{u} = (\tau_u + \tau_{d1} - m_{33}wq + m_{22}vr - d_{11}u)/m_{11} \\ \dot{v} = (\tau_{d2} - m_{11}ur - d_{22}v)/m_{22} \\ \dot{w} = (\tau_{d3} + m_{11}uq - d_{33}w)/m_{33} \\ \dot{q} = (\tau_q + \tau_{d5} + (m_{33} - m_{11})uw - WM_z\sin(\theta) - d_{55}q)/m_{55} \\ \dot{r} = (\tau_r + \tau_{d6} + (m_{11} - m_{22})uv - d_{66}r)/m_{66} \end{cases} \quad (8)$$

Where, $M = \text{diag}(m_{11}, m_{22}, m_{33}, 0, m_{55}, m_{66})$, $D(v) = \text{diag}(d_{11}, d_{22}, d_{33}, 0, d_{55}, d_{66})$, W is the buoyancy, and M_z is the metacentric height. The calculation method is shown in Eq. (9) and the rest of the parameters are constants.

$$\begin{cases} d_{11} = X_u + X_{|u|}|u| \\ d_{22} = Y_v + Y_{|v|}|v| \\ d_{33} = Z_w + Z_{|w|}|w| \\ d_{55} = M_q + M_{|q|}|q| \\ d_{66} = N_r + N_{|r|}|r| \end{cases} \quad (9)$$

The object of this paper is the American REMUS100 AUV, and based on the parameters of the REMUS-100 AUV provided in the

literature [15]. Details are given in Table 1.

3. Design of 3D trajectory tracking controller for a single underactuated AUV

When designing the 3D trajectory tracking controller, this chapter adopts the adaptive sliding mode controller, which can adjust the parameters of the sliding mode surface adaptively through the introduction of adaptive technology, so as to better adapt to the dynamic characteristics and uncertainties of the system, even when there are uncertainties and disturbances in the system. The adaptive sliding mode controller can also adjust the sliding mode surface parameters according to the real-time changes of the system state and error, so that the system has higher stability and robustness.

Therefore, the adaptive sliding mode controller has strong anti-interference characteristics and can play an important role in practical problems such as trajectory tracking. In order to achieve accurate tracking of the three-dimensional curve in space, it is necessary to ensure that the error between the position coordinates of the AUV and the position coordinates of the curve in the process of motion is within a range, so the error calculation formula is defined as follows:

$$\begin{cases} x_e = x - x_d \\ y_e = y - y_d \\ z_e = z - z_d \end{cases} \quad (10)$$

Where, $\eta_1 = [x, y, z]^T$ indicates the current position of the AUV; $\eta_d = [x_d, y_d, z_d]^T$ represents the desired position of the trajectory tracked by the AUV. In order to associate the desired position η_d with the input torque τ , it is necessary to obtain the second derivative of it. According to the AUV kinematics Eq. (8), the following formula can be simplified:

$$\begin{cases} \ddot{x}_e = \dot{u}c_\psi c_\theta - u\dot{\psi}s_\psi c_\theta - u\dot{\theta}c_\psi s_\theta - \dot{v}_\psi - v\dot{\psi}c_\psi + \dot{w}s_\theta c_\psi + w\dot{\theta}c_\theta c_\psi - w\dot{\psi}s_\theta s_\psi - \ddot{x}_d \\ \ddot{y}_e = \dot{u}s_\psi c_\theta + u\dot{\psi}c_\psi c_\theta - u\dot{\theta}s_\psi s_\theta + \dot{v}_\psi - v\dot{\psi}s_\psi + \dot{w}s_\theta s_\psi + w\dot{\theta}c_\theta s_\psi + w\dot{\psi}s_\theta c_\psi - \ddot{y}_d \\ \ddot{z}_e = -\dot{u}s_\theta - u\dot{\theta}c_\theta + \dot{w}c_\theta - w\dot{\theta}s_\theta - \ddot{z}_d \end{cases} \quad (11)$$

Where s is short for sin function and c is short for cos function. Through matrix conversion, it can be sorted into the form of matrix multiplication, as follows:

$$\begin{bmatrix} \ddot{x}_e \\ \ddot{y}_e \\ \ddot{z}_e \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} + G \begin{bmatrix} \tau_u \\ \tau_q \\ \tau_r \end{bmatrix} \quad (12)$$

Where:

$$\begin{cases} F_1 = f_u c_\psi c_\theta - u r s_\psi - u q c_\psi s_\theta - \dot{v}_\psi - \frac{v r c_\psi}{c_\theta} + \dot{w} s_\theta c_\psi + w q c_\theta c_\psi - \frac{w r s_\theta s_\psi}{c_\theta} - \ddot{x}_d \\ F_2 = f_u s_\psi c_\theta + u r c_\psi - u q s_\psi s_\theta + \dot{v}_\psi - \frac{v \psi s_\psi}{c_\theta} + \dot{w} s_\theta s_\psi + w q c_\theta s_\psi + \frac{w r s_\theta c_\psi}{c_\theta} - \ddot{y}_d \\ F_3 = -f_u s_\theta - u q c_\theta + \dot{w} c_\theta - w q s_\theta - \ddot{z}_d \end{cases} \quad (13)$$

$$G = \begin{bmatrix} \frac{c_\psi c_\theta}{m_{11}} & 0 & 0 \\ \frac{s_\psi c_\theta}{m_{11}} & 0 & 0 \\ \frac{-s_\theta}{m_{11}} & 0 & 0 \end{bmatrix}$$

As can be seen from Eq. (13), G is a singular matrix, so it is impossible to obtain the relation between G and the input torque directly through the position error. In this paper, the position error is converted to velocity error and angular velocity error by introducing a small error threshold δ and using the matrix conversion method of linear algebra. According to Eq. (2), we can convert the position error into the velocity error and the angular velocity error, because the roll of the AUV is not taken into account, so the simplified equation is as follows:

Table 1
REMUS100 AUV parameters.

Parameter	value	Parameter	value	Parameter	value	Parameter	value
m_{11}	31.41kg	m_{66}	8.33kg · m ²	$Y_{\dot{\psi} \dot{\psi}}$	131kg/m	$M_{q q }$	9.4kg · m ²
m_{22}	65.98kg	$X_{\dot{u}}$	13.5kg/s	$Z_{\dot{w}}$	66.6kg/s	M_q	6.87kg · m ² /s
m_{33}	65.98kg	$X_{u u }$	3.9kg/s	$Z_{w w }$	131kg/m	$N_{r r }$	9.4kg · m ²
m_{55}	8.33kg · m ²	$Y_{\dot{v}}$	66.6kg/s	WM_z	6N · m	N_r	6.87kg · m ² /s

$$\begin{aligned}\eta_2 &= \begin{bmatrix} u \\ v \\ w \end{bmatrix} = J_1(\eta_2)^{-1} \dot{\eta}_1 = J_1(\eta_2)^{-1} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta\cos\psi & -\sin\psi & \cos\psi\sin\theta \\ \cos\theta\sin\psi & \cos\psi & \sin\theta\sin\psi \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}^{-1} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}\end{aligned}\quad (14)$$

Define a new error equation:

$$\sigma = \varepsilon - \rho \quad (15)$$

Where, ε is defined as the position error in the object coordinate system, $\rho = [\delta, 0, 0]_T$, is the error space, δ is the value extremely less than the length of AUV.

$$\begin{aligned}\varepsilon &= J_1(\eta_2)^{-1}(\eta_1 - \eta_{1d}) \\ &= J_1(\eta_2)^{-1} \begin{bmatrix} x - x_d \\ y - y_d \\ z - z_d \end{bmatrix}\end{aligned}\quad (16)$$

Taking the first derivative of $J(\eta_2)$ yields:

$$\begin{aligned}\dot{J}(\eta_2) &= \begin{bmatrix} -q\sin\theta\cos\psi - r\sin\psi & -r\cos\psi/\cos\theta & -r\sin\psi\tan\theta + q\cos\psi\cos\theta \\ -q\sin\theta\sin\psi + r\cos\psi & -r\sin\psi/\cos\theta & -q\sin\psi\cos\theta + r\cos\psi\tan\theta \\ -q\cos\theta & 0 & q\sin\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta\cos\psi & -\sin\psi & \cos\psi\sin\theta \\ \cos\theta\sin\psi & \cos\psi & \sin\theta\sin\psi \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 0 & -r & q \\ r & 0 & r\tan\theta \\ -q & -r\tan\theta & 0 \end{bmatrix} \\ &= J(\eta_2)S(\varpi)\end{aligned}\quad (17)$$

Then take the first derivative of the error Eq. (15):

$$\begin{aligned}\dot{\sigma} &= \dot{\varepsilon} - \dot{\rho} = \dot{J}_1^{-1}(\eta_1 - \eta_{1d}) + J_1^{-1}(\dot{\eta}_1 - \dot{\eta}_{1d}) \\ &= S^T(\varpi)J_1^T(\eta_1 - \eta_{1d}) + J_1^{-1}(\dot{\eta}_1 - \dot{\eta}_{1d}) \\ &= S^T(\varpi)J_1^{-1}(\eta_1 - \eta_{1d}) - S^T(\varpi)\rho + S^T(\varpi)\rho + J_1^{-1}(\dot{\eta}_1 - \dot{\eta}_{1d}) \\ &= S^T(\varpi)(J_1^{-1}(\eta_1 - \eta_{1d}) - \rho) - S^T(\varpi)\rho + J_1^{-1}\dot{\eta}_1 - J_1^{-1}\dot{\eta}_{1d} \\ &= S^T(\varpi)\sigma - S(\varpi)\rho + \nu_1 - J_1^{-1}\dot{\eta}_{1d} \\ &= -S(\varpi)\sigma - \begin{bmatrix} 0 & -r & q \\ r & 0 & r\tan\theta \\ -q & -r\tan\theta & 0 \end{bmatrix} \begin{bmatrix} \delta \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} u \\ v \\ w \end{bmatrix} - J_1^{-1}\dot{\eta}_{1d} \\ &= -S(\varpi)\sigma + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -\delta \\ 0 & \delta & 0 \end{bmatrix} \begin{bmatrix} u \\ q \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ v \\ w \end{bmatrix} - J_1^{-1}\dot{\eta}_{1d} \\ &= -S(\varpi)\sigma + P \begin{bmatrix} u \\ q \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ v \\ w \end{bmatrix} - J_1^{-1}\dot{\eta}_{1d}\end{aligned}\quad (18)$$

Then the backstepping method is used to design the desired velocity:

$$\begin{bmatrix} u_d \\ q_d \\ r_d \end{bmatrix} = P^{-1} \left(-K\sigma - \begin{bmatrix} 0 \\ v \\ w \end{bmatrix} + J_1^{-1}\dot{\eta}_{1d} \right) \quad (19)$$

Where $K = kI$, $k > 0$, $I \in \mathbb{R}^{3 \times 3}$.

$$\dot{\sigma} = -S(\varpi)\sigma + P \begin{bmatrix} u \\ q \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ v \\ w \end{bmatrix} - P \begin{bmatrix} u_d \\ q_d \\ r_d \end{bmatrix} - \begin{bmatrix} 0 \\ v \\ w \end{bmatrix} = -(S(\varpi) + K)\sigma + P \begin{bmatrix} u_e \\ q_e \\ r_e \end{bmatrix} \quad (20)$$

The adaptive sliding mode controller is designed to ensure that $[u_e, q_e, r_e]^T$ converges to 0, and the sliding mode surface is selected as

the linear sliding mode surface:

$$\begin{cases} S_1 = c_1 u_e \\ S_2 = c_2 q_e \\ S_3 = c_3 r_e \end{cases} \quad (21)$$

Take the derivative of the Eq. (21):

$$\begin{cases} \dot{S}_1 = c_1 \dot{u}_e = \frac{c_1}{m_{11}} (m_{22}vr - m_{33}wq - d_{11}u + \tau_u + f_1 - m_{11}\dot{u}_d) \\ \dot{S}_2 = c_2 \dot{q}_e = \frac{c_2}{m_{55}} ((m_{33} - m_{11})uw - d_{55}q - WM_z \sin\theta + \tau_q + f_5 - m_{55}\dot{q}_d) \\ \dot{S}_3 = c_3 \dot{r}_e = \frac{c_3}{m_6} ((m_{11} - m_{22})uv - d_{66}r + \tau_r + f_6 - m_{66}\dot{r}_d) \end{cases} \quad (22)$$

Using the $\tanh$ function instead of the sgn function in the isokinetic approach rate, the approach rate is:

$$\begin{cases} \dot{S}_1 = -\varepsilon_1 \tanh(S_1) - k_1 S_1 \\ \dot{S}_2 = -\varepsilon_2 \tanh(S_2) - k_2 S_2 \\ \dot{S}_3 = -\varepsilon_3 \tanh(S_3) - k_3 S_3 \end{cases} \quad (23)$$

Bring Eq. (23) into (22):

$$\begin{cases} \tau_u = -m_{22}vr + m_{33}wq + d_{11}u + m_{11}((-\varepsilon_1 \tanh(S_1) - k_1 S_1)/c_1) + \dot{u}_d \\ \tau_q = -(m_{33} - m_{11})uw + d_{55}q + WM_z \sin\theta + m_{55}((-\varepsilon_2 \tanh(S_2) - k_2 S_2)/c_2) + \dot{q}_d \\ \tau_r = -(m_{11} - m_{22})uv + d_{66}r + m_{66}((-\varepsilon_3 \tanh(S_3) - k_3 S_3)/c_3) + \dot{r}_d \end{cases} \quad (24)$$

Select Lyapunov function as follows:

$$V = \frac{1}{2}S_1^2 + \frac{1}{2}S_2^2 + \frac{1}{2}S_3^2 \quad (25)$$

Bringing Eq. (23) into (25), take the derivative of the Eq. (25):

$$\begin{aligned} \dot{V} &= -\varepsilon_1 |S_1| - k_1 S_1^2 - \varepsilon_2 |S_2| - k_2 S_2^2 - \varepsilon_3 |S_3| - k_3 S_3^2 \\ &\leq -\varepsilon_1 |S_1| - \varepsilon_2 |S_2| - \varepsilon_3 |S_3| \end{aligned} \quad (26)$$

Therefore, the sliding mode controller designed in this section is asymptotically stable in the Lyapunov sense.

4. Design of obstacle avoidance controller based on artificial potential field

Due to the collision problem of multiple AUVs in the formation process and the need to avoid obstacles when there are obstacles in the moving path, relevant algorithms need to be introduced to avoid collision. In this paper, artificial potential field method is adopted to solve the collision problem among various AUVs and the obstacle avoidance problem when encountering obstacles.

The advantage of the artificial potential field method is that it is easy to understand, easy to implement, and has a good effect in dealing with the problem of obstacle avoidance in complex environment. Compared with other obstacle avoidance methods, it does not require too much prior information, and can be adjusted adaptively according to the position and size of obstacles. Fig. 4 is a schematic diagram of the force of AUV in artificial potential field.

In Fig. 4, ρ_{OA} and ρ_{OB} respectively represent the maximum influence range of obstacle A and obstacle B. As can be seen from the figure, AUV1 is located in the influence range of obstacle A and AUV2, so AUV1 is subject to the repulsive force F_A from obstacle A and the repulsive force F_{AUV2} from AUV2, whose resultant force is expressed by F_{rep1} . AUV3 is not within the range of AUV1, AUV2, obstacle A, and obstacle B, so it does not suffer any repulsive force.

The obstacle repulsive function designed in this paper is as follows:

$$U_{orep}(X) = \begin{cases} -k_{rep}\rho_o \ln(\rho(X, X_o)) & \rho(X, X_o) \leq \rho_o \\ 0 & \rho(X, X_o) > \rho_o \end{cases} \quad (27)$$

The negative gradient of the potential Eq. (27) represents the obstacle repulsive force on AUV:

$$F_{rep}(X) = \begin{cases} \frac{k_{rep}\rho_o}{\rho(X, X_o)} \frac{\partial \rho(X, X_o)}{\partial X} & \rho(X, X_o) \leq \rho_o \\ 0 & \rho(X, X_o) > \rho_o \end{cases} \quad (28)$$

Where: $X(x, y, z)$ is the current position coordinate of AUV; $X_o(x_{obs}, y_{obs}, z_{obs})$ is the position coordinate of the obstacle; ρ_o is the maximum influence distance of the obstacle; k_{rep} is the repulsive force field coefficient of the obstacle, and $\rho(X, X_o)$ is the distance between the obstacle and AUV.

The repulsive function between AUVs designed in this paper is as follows:

$$U_{arep}(X) = \begin{cases} \sum_{j \in [1,5], i \neq j} -k_{arep}\rho_{auv} \ln(\rho(X_i, X_j)) & \rho(X_i, X_j) \leq \rho_{auv} \\ 0 & \rho(X_i, X_j) > \rho_{auv} \end{cases} \quad (29)$$

The negative gradient of the potential Eq. (29) represents the repulsive forces of the AUV on other AUVs:

$$F_{arep}(X) = \begin{cases} \sum_{j \in [1,5], i \neq j} \frac{k_{arep}\rho_{auv}}{\rho(X_i, X_j)} \frac{\partial \rho(X_i, X_j)}{\partial X_i} & \rho(X_i, X_j) \leq \rho_{auv} \\ 0 & \rho(X_i, X_j) > \rho_{auv} \end{cases} \quad (30)$$

Where: $X_i(x_i, y_i, z_i)$ is the position coordinate of the i th AUV; $X_j(x_j, y_j, z_j)$ is the position coordinate of the j -th AUV; ρ_{auv} is the maximum influence distance of AUV; k_{arep} is the repulsive field coefficient of AUV, and $\rho(X_i, X_j)$ is the linear distance between the i th AUV and the j -th AUV.

In order to apply repulsive force to AUV system, it needs to be expressed in object coordinate system $\{O_A\}$, Suppose the repulsive force is: $F_{rep} = (F_{Ax}, F_{Ay}, F_{Az})$:

$$\begin{bmatrix} F_{Bx} \\ F_{By} \\ F_{Bz} \end{bmatrix} = J_1(\eta_2)^T \begin{bmatrix} F_{Ax} \\ F_{Ay} \\ F_{Az} \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ -\sin\psi & \cos\psi & 0 \\ \cos\psi\sin\theta & \sin\theta\sin\psi & \cos\theta \end{bmatrix} \begin{bmatrix} F_{Ax} \\ F_{Ay} \\ F_{Az} \end{bmatrix} \quad (31)$$

The following formula can be solved:

$$\begin{cases} F_{Bx} = F_{Ax}\cos\theta\cos\psi + F_{Ay}\cos\theta\sin\psi - F_{Az}\sin\theta \\ F_{By} = -F_{Ax}\sin\psi + F_{Ay}\cos\psi \\ F_{Bz} = F_{Ax}\cos\psi\sin\theta + F_{Ay}\sin\theta\sin\psi + F_{Az}\cos\theta \end{cases} \quad (32)$$

Feedback the repulsive force into the expected value of the sliding mode controller:

$$\begin{cases} u_{bi} = k_u(F_{Axi}\cos\theta_i\cos\psi_i + F_{Ayi}\cos\theta_i\sin\psi_i - F_{Azi}\sin\theta_i) \\ q_{bi} = k_q(-F_{Axi}\sin\psi_i + F_{Ayi}\cos\psi_i) \\ r_{bi} = k_r(F_{Axi}\cos\psi_i\sin\theta_i + F_{Ayi}\sin\theta_i\sin\psi_i + F_{Azi}\cos\theta_i) \end{cases} \quad (33)$$

Where, $k_u > 0$, $k_q > 0$, $k_r > 0$, which is defined as the correction coefficient. Combined with Eq. (19), the revised expected speed is as follows:

$$\begin{bmatrix} u_d \\ q_d \\ r_d \end{bmatrix} = P^{-1} \left(-K\sigma - \begin{bmatrix} 0 \\ v \\ w \end{bmatrix} + J_1^{-1} \dot{\eta}_{1d} \right) + \begin{bmatrix} u_{bi} \\ q_{bi} \\ r_{bi} \end{bmatrix} \quad (34)$$

5. Design of virtual structure cluster controller

The common AUV formation has been introduced in the introduction, and now the virtual structure method will be used to control the formation. The basic principle of this method is that the motion trajectories of virtual structure points are generated according to the expected trajectories and tracked by the trajectory tracking algorithm. The following AUV deduces its tracking trajectory according to the virtual structure, so as to realize formation sailing. Therefore, the virtual structure formation problem can be transformed into multiple AUV trajectory tracking problem, but the expected trajectory of the virtual structure points needs to be used to derive and calculate the tracking trajectory of the follower AUV that meets the formation requirements. This method belongs to decentralized control, and each AUV bears the corresponding calculation amount. The following will be derived according to different formations. In this paper, we assume that the virtual expected path is known, and derive other virtual structure points that the AUV needs to track through the spatial geometry.

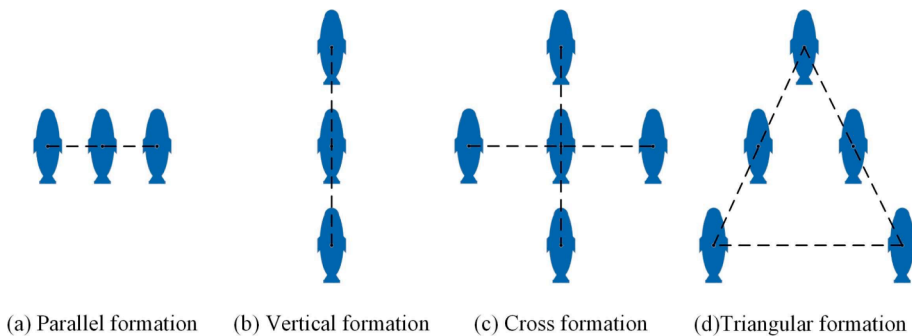


Fig. 1. The common AUV cluster formation.

Since the expected path of the virtual AUV is known, the expected path of the virtual structure points can be derived according to the structural characteristics of the parallel formation. The derivation of the parallel formation in Fig. 1 is shown in Fig. 5.

P_0 is the structure point of the virtual expected trajectory., and u_d is the longitudinal velocity direction of the virtual AUV. In order for the AUV formation to be parallel to the virtual AUV, the line between the virtual structure point of the AUV and the virtual AUV must be perpendicular to the velocity direction of the virtual AUV. Therefore, the plane equation perpendicular to the velocity u_d of the virtual underwater vehicle can be solved by the theorem in advanced mathematics.

The desired position of the AUV is defined as (x, y, z) . Therefore, based on the desired position $P_0(x_d, y_d, z_d)$ of the virtual AUV and the linear velocity direction $u_d(dx_d, dy_d, dz_d)$, the equation for this plane can be derived:

$$dx_d(x - x_d) + dy_d(y - y_d) + dz_d(z - z_d) = 0 \quad (35)$$

The spacing of each AUV is now defined as R . Then the expected position of the central AUV is located on the sphere with point P_0 as the center of the circle and radius is R . So the spherical equation is as follows:

$$(x - x_d)^2 + (y - y_d)^2 + (z - z_d)^2 = R^2 \quad (36)$$

All the points satisfying the requirement can be solved by combining the formula (35) and (36), which are located on the circle M in the Fig. 5. Since AUVs should be arranged in a fixed parallel formation, the point that is the same as the desired coordinate point Z coordinate of the virtual leader is selected as the desired position coordinate of the follower AUVs.

$$z = z_d \quad (37)$$

The expected path of the AUV of the parallel formation can be solved by Eqs. (35) and (36) and (37). The above method is also adopted for vertical formation. It is only necessary to combine the spherical equation and the path of virtual AUV. The derivation process of virtual structure points is shown in Fig. 6.

The expected paths of all AUVs over time can be obtained by curve fitting. The cross formation is the combination solution of parallel formation and vertical formation, and the path derivation can be seen in Fig. 7.

For triangular formation, the curve $(l_1, l_2, l_3, l_4, l_5)$ with interval d can be solved by parallel formation method, and the expected path point of virtual AUV is taken as the center of the circle, and the hemispherical surface can be made by R and $2R$ respectively. Its intersection with the curve is the expected path point following the AUV.

It is difficult to express the hemispherical equation when solving the expected path of the following AUV for triangular formation. We can choose by judging the positive or negative cosine of the angle between the virtual AUV and the following AUV. The cosine behind the virtual AUV is negative and the cosine in front is positive. Please refer to Figs. 8 and 9 for details. where $\cos\theta_1 > 0$, $\cos\theta_2 < 0$. But the premise is $d < R$.

6. Simulation experiment of AUV cluster formation obstacle avoidance

In order to verify the effectiveness of the proposed algorithm, simulation experiments are designed to verify it. Firstly, it is tested whether the artificial potential field method can effectively avoid collision between AUVs. Secondly, the simulation of five formation obstacle avoidance types parallel formation, parallel formation, cross formation and triangular formation are carried out.

6.1. Obstacle avoidance experiment between AUVs

In order to verify that the artificial potential field obstacle avoidance algorithm designed in Chapter 2 is effective for obstacle avoidance between AUVs, two AUVs are allowed to track two inevitably intersecting tracking paths respectively to verify the effectiveness of the algorithm. The intersection points of these two tracking paths are synchronized in time so that the two AUVs tracking this curve will inevitably collide. Details of simulation parameters are shown in Table 2.

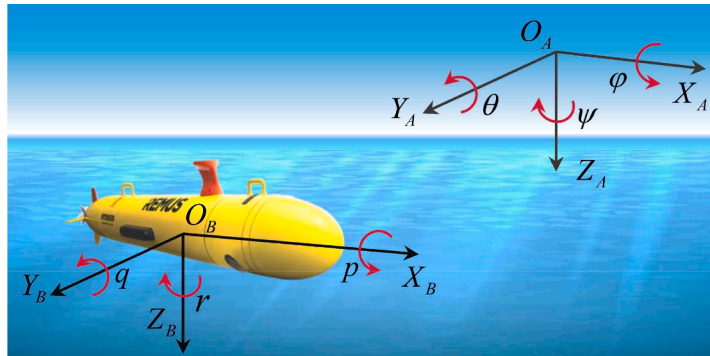


Fig. 2. Coordinate system definition of AUV.

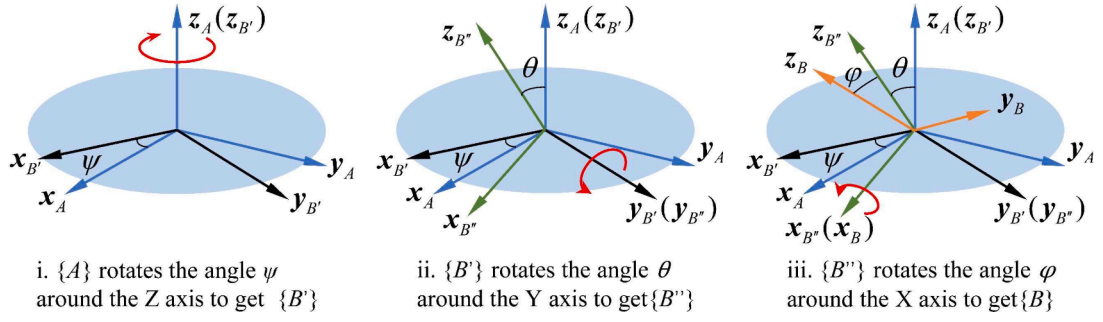


Fig. 3. Fixed coordinate system and object coordinate system coordinate conversion diagram.

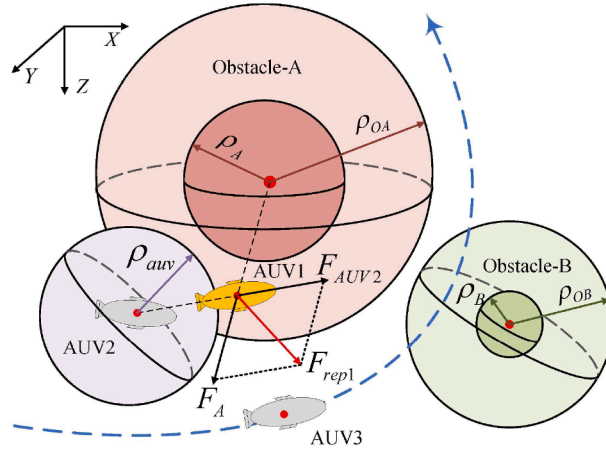


Fig. 4. AUV force diagram under artificial potential field method.

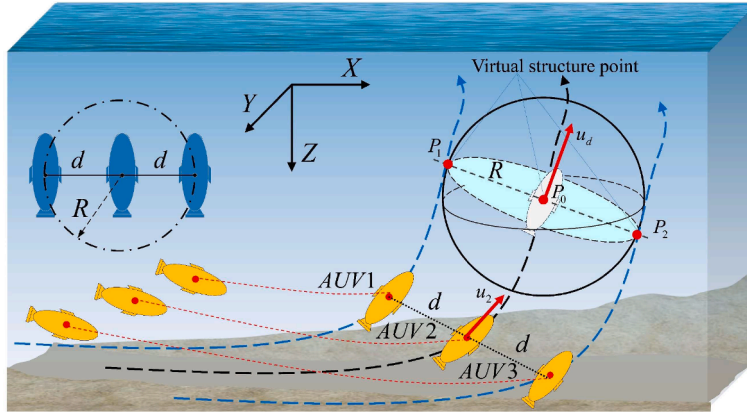


Fig. 5. The expected path derivation of parallel formation AUV.

Parametric equation for line 1 and line 2:

$$\text{Line1: } \begin{cases} x_{d1} = 3t \\ y_{d1} = 20 + t \\ z_{d1} = 10 + 2t \end{cases} \quad \text{Line2: } \begin{cases} x_{d2} = 10 + 2t \\ y_{d2} = 20 + t \\ z_{d2} = 3t \end{cases} \quad (38)$$

The experimental results of obstacle avoidance between AUVs verified by artificial potential field method are as follows. By comparing Fig a and Fig b in Fig. 10, it can be seen that the AUV tracking path under the action of artificial potential field method has a

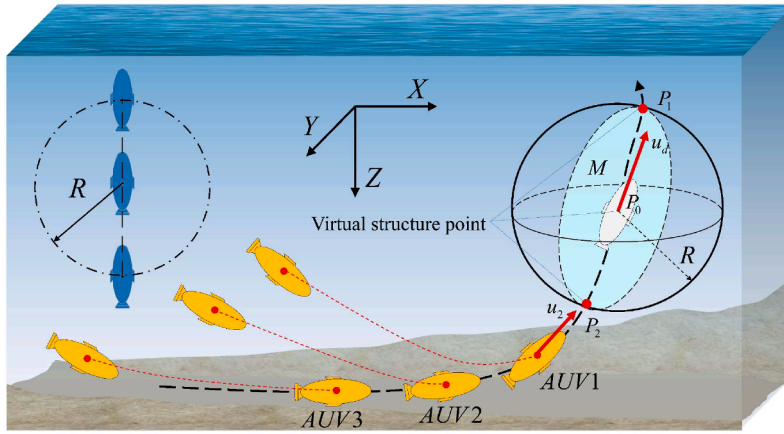


Fig. 6. The expected path derivation of vertical formation AUV.

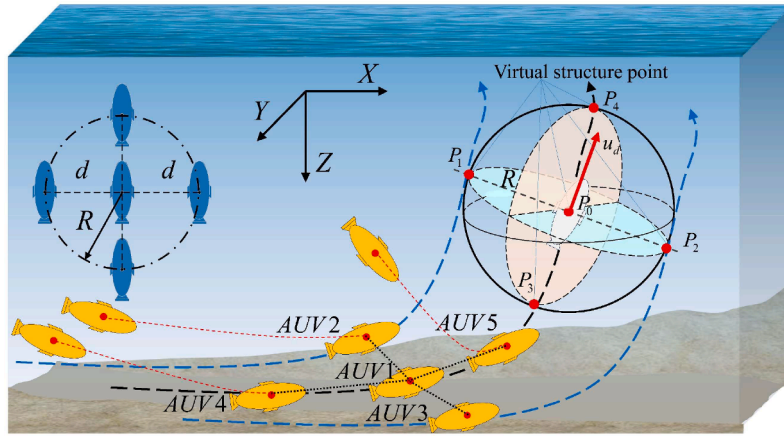


Fig. 7. The expected path derivation of cross formation AUV.

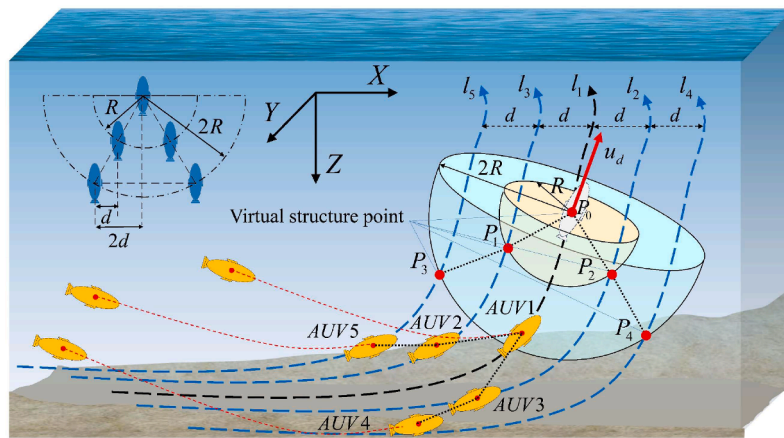


Fig. 8. The expected path derivation of triangular formation AUV.

significant deviation, indicating that artificial potential field plays a role in obstacle avoidance.

Fig. 11 shows the distance curve between two AUVs. It can be seen that under the action of artificial potential field, the minimum distance between two AUVs is not 0, and its set value is $\rho_{aav} = 3\text{m}$. It is proved that the artificial potential field designed in this paper

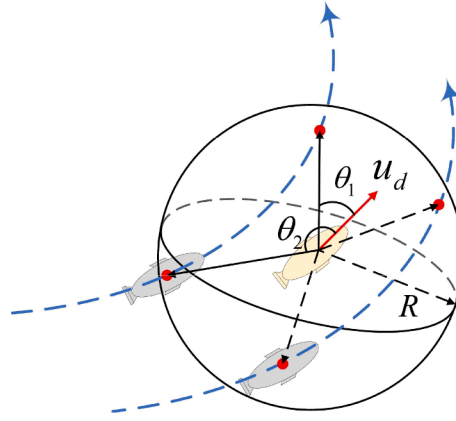


Fig. 9. The method of solving the expected point of AUV by triangular formation.

Table 2

Experimental parameters of collision between two AUVs.

Parameter	value	Parameter	value	Parameter	value
ρ_{aav}	3m	η_1 : Initial position of AUV 1	[- 50, 50, 0, 0, 0, 0]	v_1 : Initial speed of AUV 1	[0, 0, 0, 0, 0, 0]
k_{arep}	1000	η_2 : Initial position of AUV 2	[- 50, -50, 0, 0, 0, 0]	v_2 : Initial speed of AUV 2	[0, 0, 0, 0, 0, 0]
δ	0.1	$[k_w, k_p, k_r]$	[0.1, 0.01, 0.01]	Simulated step size	0.001s
k	1	$[c_1, c_2, c_3]$	[8, 10, 10]	$[k_1 k_2 k_3]$	[20, 25, 25]
		$[\epsilon_1, \epsilon_2, \epsilon_3]$	[1, 1, 1]	t	10s

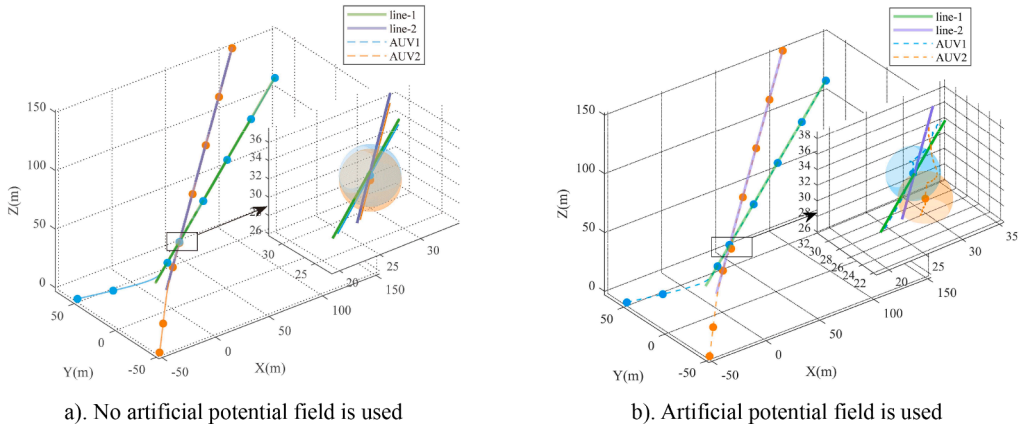


Fig. 10. Comparison diagram under the action of artificial potential field.

can effectively avoid collision between AUVs.

6.2. Simulation experiment of four kinds of formation obstacle avoidance

The simulation parameters of the four formation obstacle avoidance are shown in Table 3. According to Fig. 5, the key index of the position solution element of the parallel formation can be obtained as R_1 . Similar, according to Fig. 6, the key index of the position solution element of the vertical formation can be obtained as R_2 . According to Fig. 7, the key index of the position solution element of the cross formation can be obtained as R_3 . According to Fig. 8, the key index of the position solution element of the triangular formation can be obtained as R_4 .

Expected tracking path:

$$\begin{cases} x_d = 100\sin(0.1t) \\ y_d = 100\cos(0.1t) \\ z_d = t \end{cases} \quad (39)$$

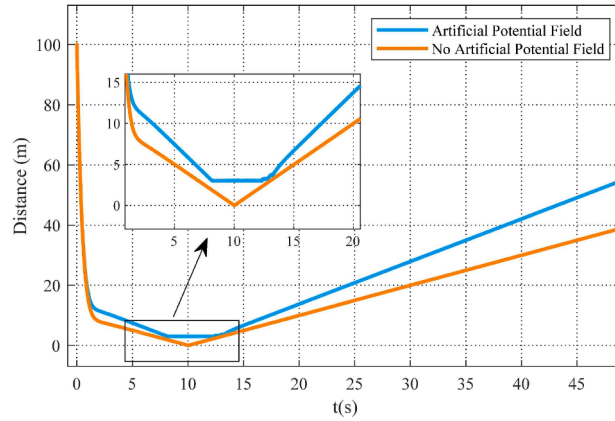


Fig. 11. Comparison diagram under the action of artificial potential field.

Table 3

Experimental parameters of four kinds of formation obstacle avoidance.

Parameter	value	Parameter	value	Parameter	value
ρ_{auv}	3m	η_1 : Initial position of AUV 1	[0, 0, 0, 0, 0]	v_1 : Initial speed of AUV 1	[0, 0, 0, 0, 0]
ρ_o	10m	η_2 : Initial position of AUV 2	[-10, -10, -10, 0, 0]	v_2 : Initial speed of AUV 2	[0, 0, 0, 0, 0]
ρ_{obs}	3m	η_3 : Initial position of AUV 3	[-20, 100, -10, 0, 0]	v_3 : Initial speed of AUV 3	[0, 0, 0, 0, 0]
k_{arep}	1000	η_4 : Initial position of AUV 4	[-10, -100, -10, 0, 0]	v_4 : Initial speed of AUV 4	[0, 0, 0, 0, 0]
k_{rep}	100	η_5 : Initial position of AUV 5	[20, 80, 0, 0, 0]	v_5 : Initial speed of AUV 5	[0, 0, 0, 0, 0]
δ	0.1	obs1: Coordinates for obstacle1	[18.5, -103, 29.5]	$[k_{ip}, k_p, k_r]$	[0.1, 0.01, 0.01]
k	1	obs2: Coordinates for obstacle2	[85, 12, 14]	$[c_1, c_2, c_3]$	[8, 10, 10]
		obs3: Coordinates for obstacle3	[-115, -9, 46]	$[e_1, e_2, e_3]$	[1, 1, 1]
		τ_{d1} : u - disturbance	$20 + 100\sin(0.02t)$	$[k_1, k_2, k_3]$	[20, 25, 25]
		τ_{d2} : v - disturbance	$15 + 100\sin(0.02t)$	Parallel formation: R_1	10m
		τ_{d3} : w - disturbance	$10 + 100\sin(0.02t)$	Vertical formation: R_2	10m
		τ_{d5} : q - disturbance	$5 + 100\sin(0.01t)$	Cross formation: R_3	10m
		τ_{d6} : r - disturbance	$5 + 100\sin(0.02t)$	Triangular formation: R_4	15m
		Simulated step size	0.001s	Total simulation time	50s

Table 4

Experimental parameters of formation experiment in complex environment.

Parameter	value	Parameter	value	Parameter	value
ρ_{auv}	3m	η_1 : Initial position of AUV 1	[0, 0, 0, 0, 0]	v_1 : Initial speed of AUV 1	[0, 0, 0, 0, 0]
ρ_{TC}	10m	η_2 : Initial position of AUV 2	[-10, -10, -10, 0, 0]	v_2 : Initial speed of AUV 2	[0, 0, 0, 0, 0]
ρ_o	10m	η_3 : Initial position of AUV 3	[-20, 100, -10, 0, 0]	v_3 : Initial speed of AUV 3	[0, 0, 0, 0, 0]
ρ_M	20m	η_4 : Initial position of AUV 4	[-10, -100, -10, 0, 0]	v_4 : Initial speed of AUV 4	[0, 0, 0, 0, 0]
k_{arep}	1000	η_5 : Initial position of AUV 5	[20, 80, 0, 0, 0]	v_5 : Initial speed of AUV 5	[0, 0, 0, 0, 0]
k_{rep}	100	η_6 : Initial position of AUV 4	[-30, 120, 0, 0, 0]	v_6 : Initial speed of AUV 4	[0, 0, 0, 0, 0]
δ	0.1	η_7 : Initial position of AUV 5	[20, 20, 0, 0, 0]	v_7 : Initial speed of AUV 5	[0, 0, 0, 0, 0]
k	1	obs1: Coordinates for obstacle1	[18.5, -103, 29.5]	obs5: obstacle_Max1	[-60, -15, 42]
		obs2: Coordinates for obstacle2	[85, 12, 14]	obs6: obstacle_Max 2	[72, -103, 22]
		obs3: Coordinates for obstacle3	[-115, -9, 46]	$[k_{ip}, k_p, k_r]$	[0.1, 0.01, 0.01]
		obs4: Coordinates for obstacle1	[-70.7, -107, 36.5]	$[c_1, c_2, c_3]$	[8, 10, 10]
		τ_{d1} : u - disturbance	$20 + 100\sin(0.02t)$	$[e_1, e_2, e_3]$	[1, 1, 1]
		τ_{d2} : v - disturbance	$15 + 100\sin(0.02t)$	$[k_1, k_2, k_3]$	[20, 25, 25]
		τ_{d3} : w - disturbance	$10 + 100\sin(0.02t)$	Simulated step size	0.001s
		τ_{d5} : q - disturbance	$5 + 100\sin(0.01t)$	Total simulation time	50s
		τ_{d6} : r - disturbance	$5 + 100\sin(0.02t)$	Parallel formation: R_1	10m

According to the above simulation parameters and the expected tracking path, the following simulation results can be obtained. Fig. 12 shows the representation of the AUV and obstacle in the simulation diagram, and their sphere of influence of the artificial potential field repulsion is represented by a transparent sphere. ρ_o represents the influence range of obstacles, ρ_{obs} represents the actual size of the obstacle, and ρ_{auv} represents the influence range of AUV.

The simulation results of parallel formation are shown in Figs. 13 and 14.

The simulation results of vertical formation are shown in Figs. 15 and 16.

Table 5

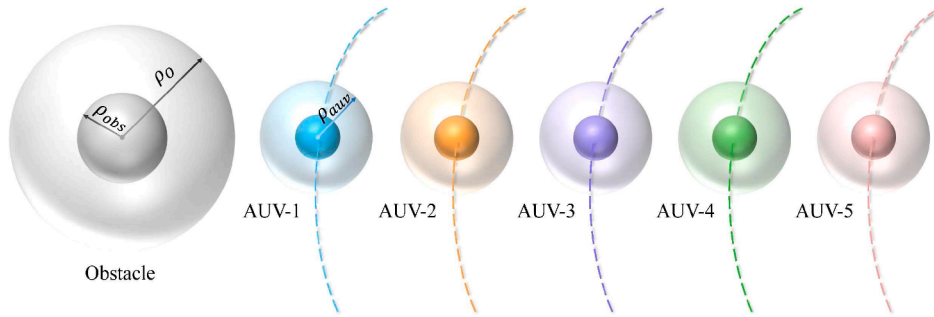
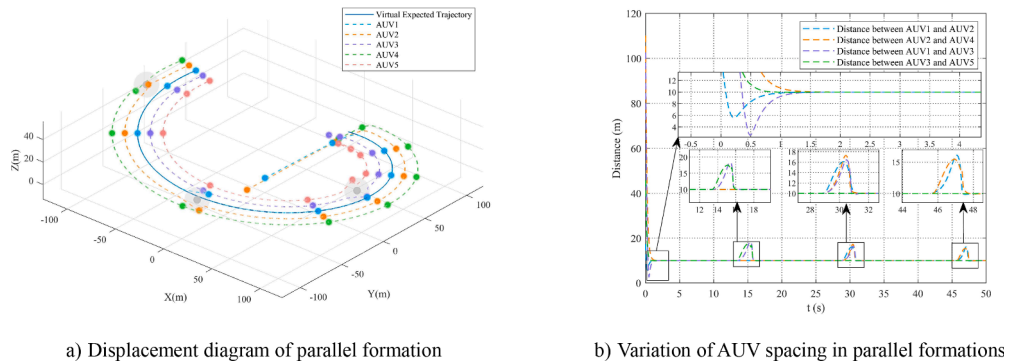
Experimental parameters of formation algorithm extensibility verification.

Parameter	value	Parameter	value	Parameter	value
δ	0.1	η_x : Initial position range in the X direction of AUV	[100, -100]	v_x : Initial speed of AUVs	[0, 0, 0, 0, 0]
k	1	η_y : Initial position range in the Y direction of AUV	[100, -100]	η_s : Initial space attitude of AUVs	[0, 0, 0]
		η_z : Initial position range in the Z direction of AUV	[50, -50]	Increasing rule of AUV number	3,5,7,9,11...99
		τ_{d1} : u - disturbance	$25 + 100\sin(0.02t)$	$[c_1, c_2, c_3]$	[8, 10, 10]
		τ_{d2} : v - disturbance	$25 + 100\sin(0.02t)$	$[\varepsilon_1, \varepsilon_2, \varepsilon_3]$	[1, 1, 1]
		τ_{d3} : w - disturbance	$25 + 100\sin(0.02t)$	$[k_1 k_2 k_3]$	[20, 25, 25]
		τ_{d5} : q - disturbance	$15 + 100\sin(0.01t)$	Simulated step size	0.001s
		τ_{d6} : r - disturbance	$15 + 100\sin(0.02t)$	Total simulation time	50s
		Parallel formation: R_1	10m	Cross formation: R_3	10m
		Vertical formation: R_2	10m	Triangular formation: R_4	15m

Table 6

Experimental parameters of formation algorithm extensibility verification.

Parameter	value	Parameter	value
Controller gain α	0.8	Cost function coefficient Q_1	0.01
Controller gain β	2.5	Cost function coefficient Q_2	0.01
τ_{d1} : u - disturbance	$50 + 100\sin(0.02t)$	Time varying current disturbance	[0, 0, 0, 0, 0]
τ_{d2} : v - disturbance	$50 + 100\sin(0.02t)$	Trajectory in the X direction	$X_d = 100\sin(0.1t)$
τ_{d3} : w - disturbance	$50 + 100\sin(0.02t)$	Trajectory in the Y direction	$Y_d = 100\cos(0.1t)$
τ_{d5} : q - disturbance	$10 + 100\sin(0.01t)$	Trajectory in the Z direction	$Z_d = t$
τ_{d6} : r - disturbance	$10 + 100\sin(0.02t)$	Total simulation time	50s

**Fig. 12.** Description of obstacles and AUV legend in simulation.**Fig. 13.** Formation details for parallel formation.

The simulation results of corss formation are shown in Figs. 17 and 18.

The simulation results of triangular formation are shown in Figs. 19 and 20.

From the formation path diagram as shown in Fig. 13(a), Fig. 15(a), Fig. 17(a) and Fig. 19(a), it can be seen that the profile of each AUV formation basically meets the requirements of parallel formation, vertical formation, cross formation and triangular formation.

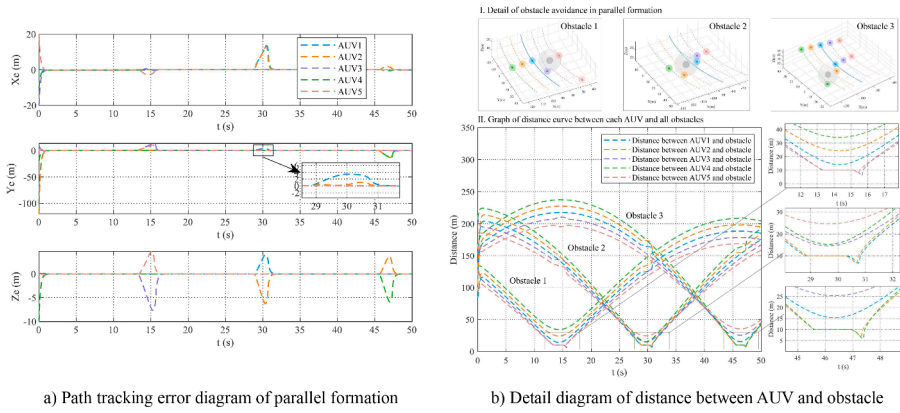


Fig. 14. Obstacle avoidance details for parallel formation.

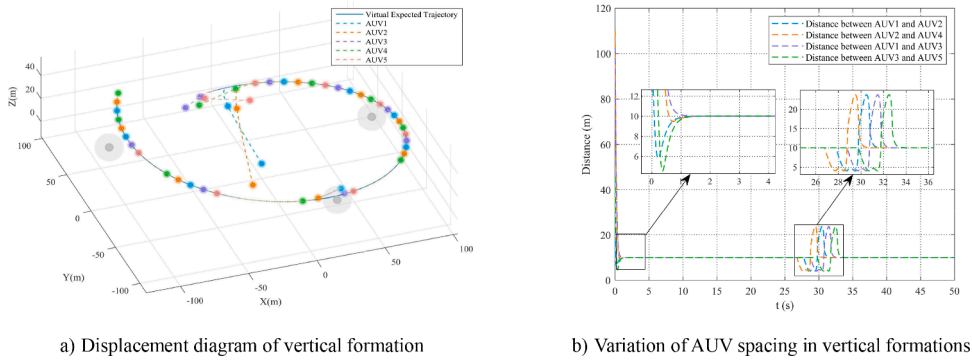


Fig. 15. Formation details for vertical formation.

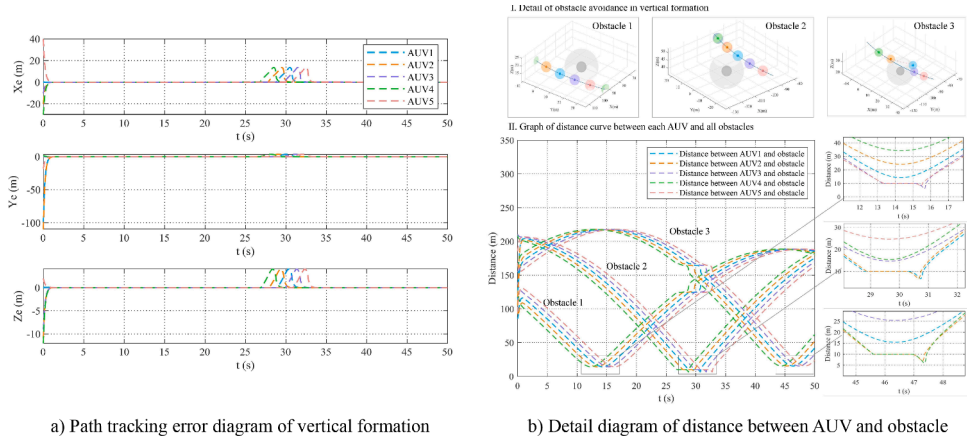


Fig. 16. Obstacle avoidance details for vertical formation.

By observing the AUV spacing details of each formation, as shown in Fig. 13(b), Fig. 15(b), Fig. 17(b) and Fig. 19(b), it can be found that the key elements of the formation of each formation are basically stable at the set value of 10, 10, 10, 15 m, and there is an obvious position deviation in the obstacle area. By observing the AUV tracking error diagram of each formation, as shown in Fig. 14(a), Fig. 16(a), Fig. 18(a) and Fig. 20(a), it can be seen that the tracking error of each AUV, e_x, e_y, e_z , basically converges to 0 when it does not enter the range of obstacles, and obvious error deviation can be generated in the area of obstacles. By observing the obstacle avoidance details of each formation, as shown in Fig. 14(b), Fig. 16(b), Fig. 18(b) and Fig. 20(b), it can be found that when AUV reaches the influence range of the obstacle, the distance between AUV and the obstacle will change significantly and remain near the influence range of the obstacle. This shows that the AUV in the influence range of obstacles is affected by the repulsive force from the artificial

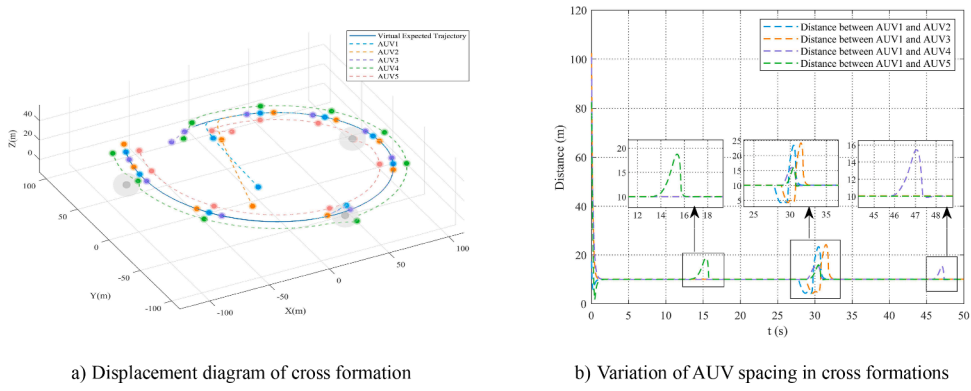


Fig. 17. Formation details for cross formation.

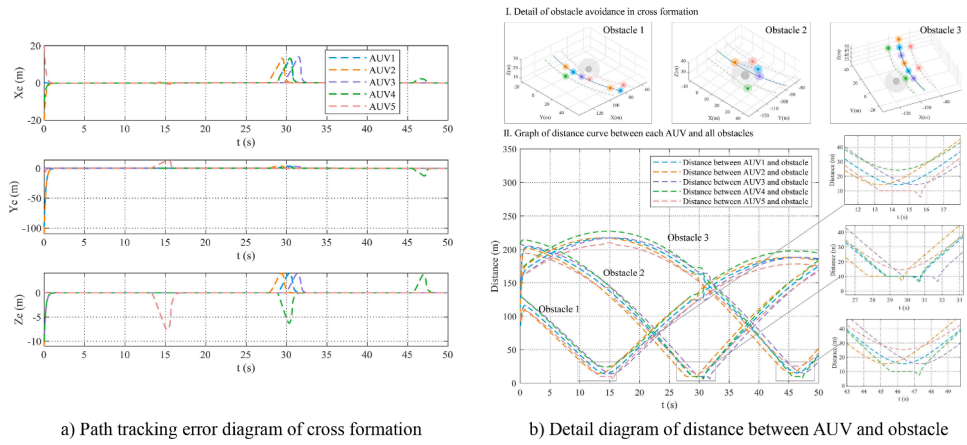


Fig. 18. Obstacle avoidance details for cross formation.

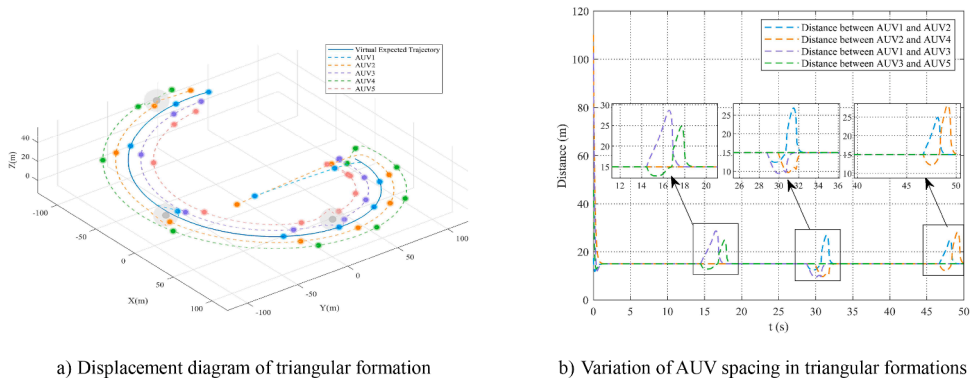


Fig. 19. Formation details for triangular formation.

potential field method of obstacles, and changes the original tracking trajectory. It can be seen that the obstacle avoidance of the artificial potential field method is effective by observing the obstacle avoidance detail diagrams in each diagram.

6.3. Simulation of multiple AUVs in formation in unpredictable and complex environments

The main objective of this experiment is to explore the simulation of formation control of multiple AUVs in unpredictable and complex environments. The simulation environment is constructed by considering the size and location of obstacles in the environment and the detection range of AUV. As shown in Fig. 21, ρ_M represents the influence range of large obstacles, ρ_{Mobs} represents the actual

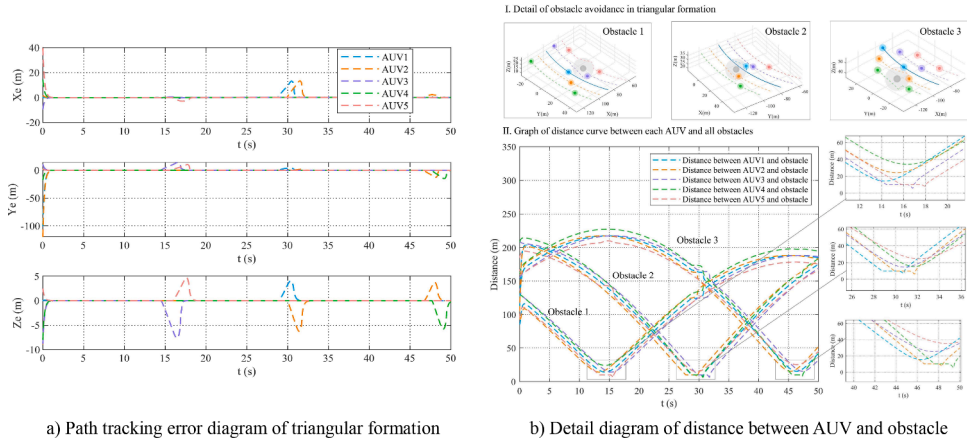


Fig. 20. Obstacle avoidance details for triangular formation.

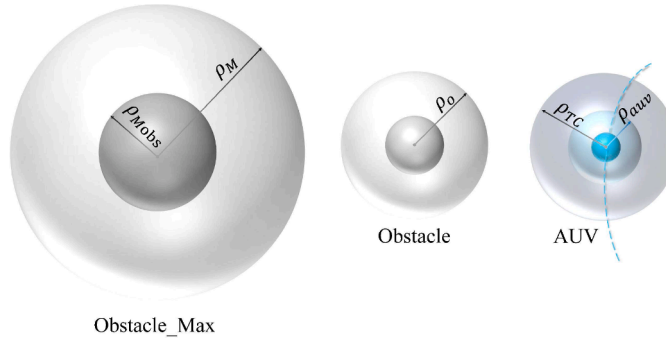


Fig. 21. Description of different obstacles and AUV detection range.

size of the obstacle, and ρ_{TC} represents the detection range of AUV, within which AUV can perceive the position information of obstacles. The expected tracking curve remains unchanged, consistent with the simulation experiment in the previous section.

In this simulation experiment, 7 AUVs were used for formation simulation, and the simulation formation was parallel formation. Specific simulation parameters are shown in Table 4.

The simulation results of parallel formation are shown in Fig. 22 Figs. 23 and 24.

As can be seen from the formation path diagram shown in Fig. 22, the formation of 7 AUVs basically meets the requirements of

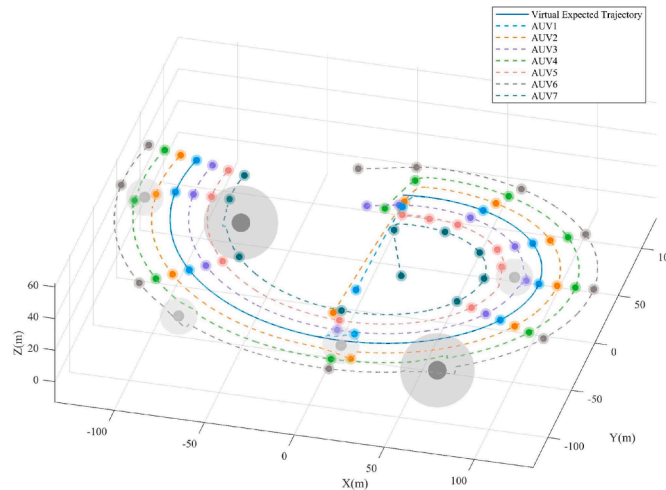


Fig. 22. Simulation of formation obstacle avoidance in complex environment.

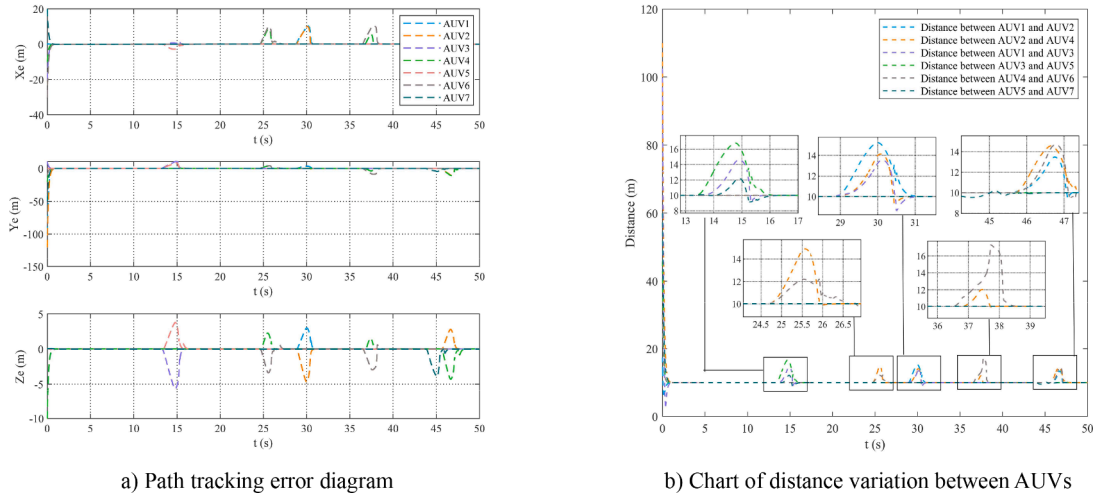


Fig. 23. Simulation details in complex environments.

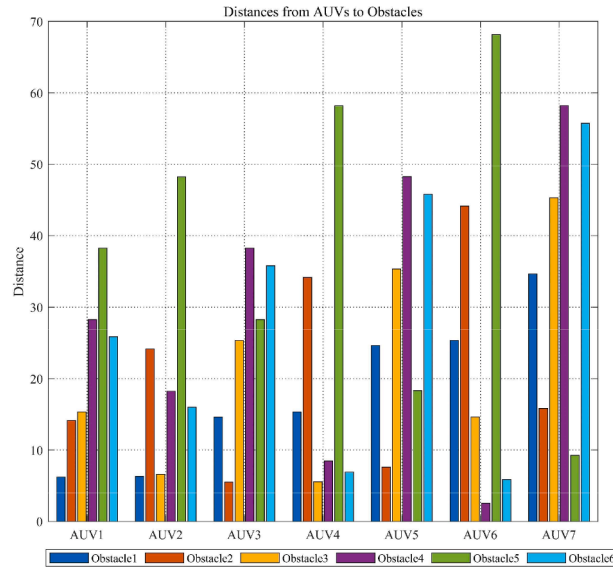


Fig. 24. Detail diagram of distance between AUV and obstacle.

parallel formation. By observing the detail diagram of the spacing between AUVs in Fig. 23, it can be found that the key elements of parallel formation are basically stable at the set value of 10 m, and there are obvious position deviations in the area of obstacles and large obstacles. By observing the tracking error diagram of parallel formation AUVs as shown in Fig. 23(a), it can be seen that the tracking errors e_x , e_y and e_z of each AUV basically converge to 0 when they do not enter the range of obstacles, and obvious position deviation will occur in the area of obstacles. According to the observation in Fig. 24, the distance between each AUV and all obstacles is within the safe range in a complex environment and considering the detection range of AUV, which proves the effectiveness of the artificial potential field method designed in this paper.

6.4. Simulation experiment of extensibility of formation obstacle avoidance algorithm and comparison verification experiment

In this section, we will further explore the scalability of the algorithm through simulation experiments. Specifically, we will examine how the algorithm performs in formation with different numbers of AUVs. By studying the change of formation solution time growth rate and path tracking error with each additional pair of AUVs, the scalability of formation algorithm is verified. Due to the symmetry of the formations designed in this paper, the number of AUVs will increase according to the law of 3, 5, 7, 9 and 11. Through this experiment, we aim to deeply explore the scalability of the algorithm in different formations, and provide guidance for future practical applications. Experimental simulation parameters are shown in Table 5.

Considering that the turning radius of the tracking curve of the innermost AUV is too small due to the enlargement of the AUV formation, the following expected trajectory is used for calculation:

$$\begin{cases} x_d = 1000\sin(0.1t) \\ y_d = 1000\cos(0.1t) \\ z_d = 10t \end{cases} \quad (40)$$

Simulation data are shown in the figure below:

By observing Fig. 25, it can be found that as the number of AUVs in formation increases in pairs, the increase time of the calculation time of the four formation algorithms, parallel formation, vertical formation, cross formation and triangle formation, is stable at 0.4749, 0.2986, 1.2291 and 2.2982 s respectively. By observing the average tracking error of each axis of AUV of different formations, it can be found that the error values are kept between 0.003 and 0.004 m. Through experiments, it can be found that when the number of AUVs increases in pairs, the computational growth rate of the formation algorithm in this paper remains stable, and the tracking error can be maintained at a stable small value when the number of AUVs increases. This experiment can prove that the formation algorithm in this paper is scalable.

Empirical verification is the key link to ensure that theory conforms to practice. We will conduct a comprehensive test and evaluation of the algorithm in this paper and the algorithm in reference [21]. This reference mainly adopts the formation algorithm of virtual leader, and studies the problem of multi-AUV cost-preserving formation tracking control based on switching topology communication, and realizes the formation tracking control of multi-AUV system. This section mainly compares the performance of the parallel formation under the same parameters. This can not only verify the practicability and reliability of the algorithm, but also provide valuable data support and theoretical basis for the subsequent optimization iteration and practical application.

The parameter Settings in this paper are consistent with those in parallel formation simulation experiment in 6.2. The parameter Settings in reference [21] are shown in Table 6.

Simulation data are shown in Fig. 26:

In the following table, Virtual leader represents the method used in the reference [21], Virtual structure represents the algorithm used in this paper. Comparison table of key performance parameters of the two algorithms:

In Table 7, Time represents the time it takes for two algorithms to run from start to finish on the same device. F_{dis} represents the formation retention degree of the parallel formation In the setting of the experiment, the key parameter of the parallel formation is the transverse distance between AUVs, which is set at 10 m. e_{x0}, e_{y0}, e_{z0} are the average values of the path tracking errors in the three directions of all AUVs respectively. Through data comparison, it can be found that under parallel formation, the calculation time of the algorithm proposed in this paper is less than that of the virtual leader algorithm in the reference literature, and it is also better than the virtual

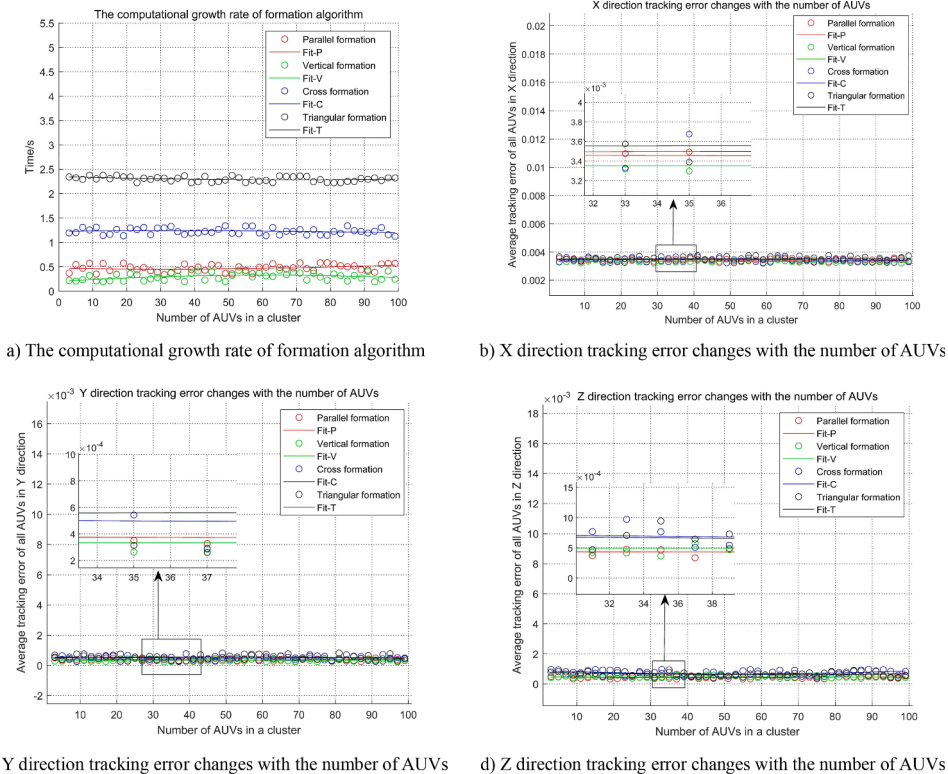


Fig. 25. Experimental verification of algorithm scalability.

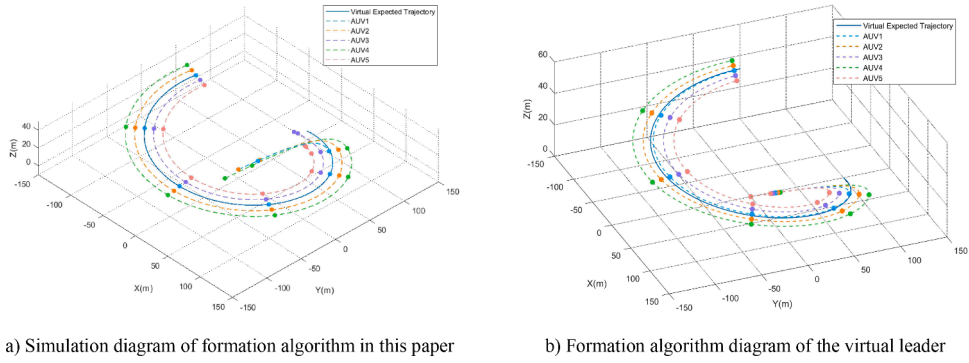


Fig. 26. Experiment of empirical verification of algorithm.

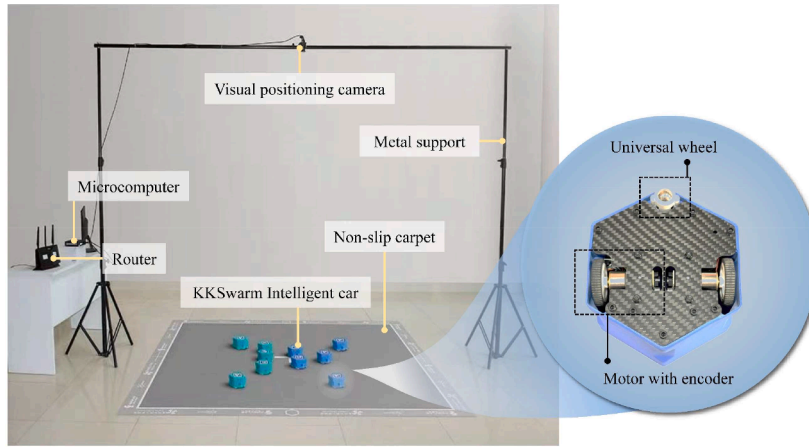


Fig. 27. KKSwarM Intelligent clustering system.

Table 7

Comparison table of key performance parameters of the two algorithms.

Index	Time	F_{dis}	e_x	e_y	e_z
virtual leader	7.1945s	8.3374	1.2085×10^{-2}	6.347×10^{-3}	4.335×10^{-3}
virtual structure	5.2886s	9.9861	2.353×10^{-3}	-9.91×10^{-4}	-1.379×10^{-3}
Extent of growth	↓26.49%	↑19.77%	↓80.53%	↓84.39%	↓68.19%

leader algorithm in formation retention degree F_{dis} . In terms of anti-interference, both algorithms are tested under the same interference environment. By observing the error values e_x, e_y, e_z , it can be found that both of them have better anti-interference performance, but the anti-interference performance of the algorithm in this paper is slightly better.

6.5. Experimental verification of formation obstacle avoidance algorithm

In order to verify the effectiveness of the formation obstacle avoidance algorithm designed in this paper in the real world, this chapter uses KKSwarM intelligent clustering system to conduct experimental verification of the algorithm. This system is used to verify the effectiveness and reliability of the proposed algorithm in two-dimensional motion. The overall structure of the system is shown in Fig. 27.

The working principle of the system is shown in the Fig. 28. KKSwarM project builds a LAN through a router, and the cluster car serves as the client. All algorithms and positioning systems are computed on the server host. Then through the TCP/IP protocol, the control command is sent to the cluster car, so as to control the movement of the car. At the same time, the host server obtains the position and direction of the cluster car through the camera in real time.

The process of this physical experiment is as follows: firstly, the experimental formation parameters and controller parameters are set; secondly, the algorithm in this paper is used to solve the state information such as path points and point speed that each unit intelligent car needs to follow, finally, the KKSwarM system carries out real-time control and reads the state information of each

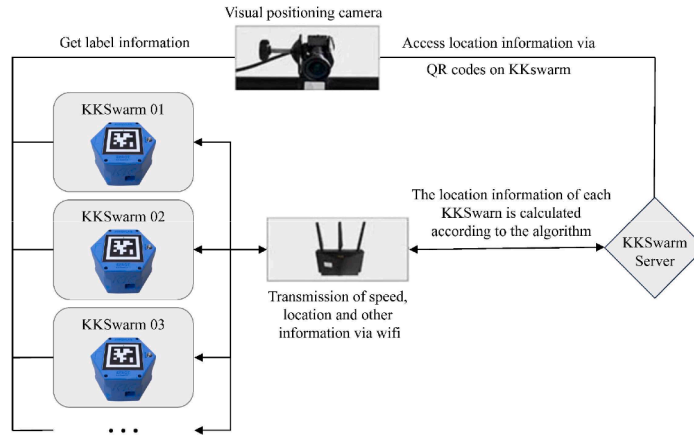


Fig. 28. KKSwarM Intelligent clustering system.

intelligent car. Finally, the experimental results are analyzed to verify the effectiveness of the algorithm.

The definition of the relevant parameters in this experiment are shown in Fig. 29. Where, l_{ob} represents the distance from the center of the obstacle to Y wheelbase, ρ_{ob1} represents the size of the virtual obstacle, and ρ_{ob2} represents the repulsive force range of the virtual obstacle. ρ_{kk} represents the repulsive range of the smart car. l_{vk} and l_{tk} respectively represent the expected trajectory of vertical formation and triangular formation from the Y wheelbase. R_{vk} and R_{tk} represent the formation elements of vertical formation and triangular formation respectively. The details of the parameters are shown in Table 8.

KKSwarM simulation results are shown in Fig. 30, Fig. 31 and Fig. 32. The intelligent vehicle displacement information and obstacle information in KKSwarM system are exported and sorted, as shown in Fig. 32. Through observation of the experimental results, it can be found that KKSwarM can better maintain formation and avoid obstacles by using the formation obstacle avoidance algorithm proposed in this paper. In Fig. 32, it is found that the intelligent car has obvious position deviation within the influence range of obstacles, which proves that it is affected by the repulsion force of obstacles, and also reflects the effectiveness of the artificial potential field algorithm proposed in this paper.

7. Conclusion

The effectiveness and scalability of the proposed formation algorithm are verified by multi-group simulation experiments and real KKSwarM experiments in two-dimensional environment. The advantages of the formation control method proposed in this paper are as follows: First, because the basic tracking route is planned off-line, the real-time requirement and calculation amount between AUVs are reduced, and the task completion is more reliable. Second, only distance feedback between AUVs is needed to avoid collisions in the

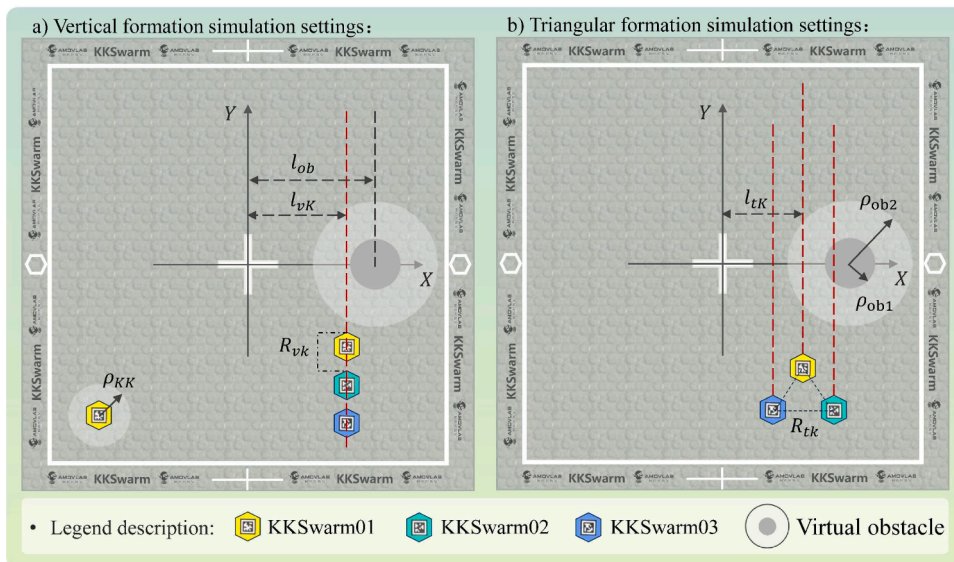


Fig. 29. KKSwarM Intelligent clustering system.

Table 8

Experimental verification parameters of formation obstacle avoidance algorithm.

Parameter	value	Parameter	value	Parameter	value
ρ_{KK}	10cm	Vertical formation		v_1 : Initial speed of KKSwarms01	[0,0]
ρ_{ob1}	8cm	η_1 : Initial position of KKSwarms01	[40, -40]	v_2 : Initial speed of KKSwarms02	[0,0]
ρ_{ob2}	28cm	η_2 : Initial position of KKSwarms02	[40, -60]	v_3 : Initial speed of KKSwarms03	[0,0]
l_{ob}	50cm	η_3 : Initial position of KKSwarms03	[40, -80]		
l_{vK}	40cm	Triangular formation			
l_{tK}	30cm	η_1 : Initial position of KKSwarms01	[35, -40]	v_1 : Initial speed of KKSwarms01	[0,0]
R_{vK}	20cm	η_2 : Initial position of KKSwarms02	[25, -57]	v_2 : Initial speed of KKSwarms02	[0,0]
R_{tK}	24cm	η_3 : Initial position of KKSwarms03	[45, -57]	v_3 : Initial speed of KKSwarms03	[0,0]
k_{arep}	1000	$[k_1 k_2 k_3]$	[20, 25, 25]	$[e_1, e_2, e_3]$	[1, 1, 1]
k_{rep}	100	$[k_{ib}, k_q, k_r]$	[0.1, 0, 0]	Calculating step size	0.001s
t	10s	$[c_1, c_2, c_3]$	[8, 10, 10]	Total simulation time	50s

**Fig. 30.** Simulation of vertical formation using KKSwarms system.**Fig. 31.** Simulation of triangular formation using KKSwarms system.

underwater environment, reducing the computational burden on each AUV robot, which all carry the corresponding computational load. Finally, the combination of this method and artificial potential field method greatly improves the obstacle avoidance performance of the algorithm.

However, the disadvantage of the proposed algorithm is that it takes a long time to solve the virtual mechanism points tracked by

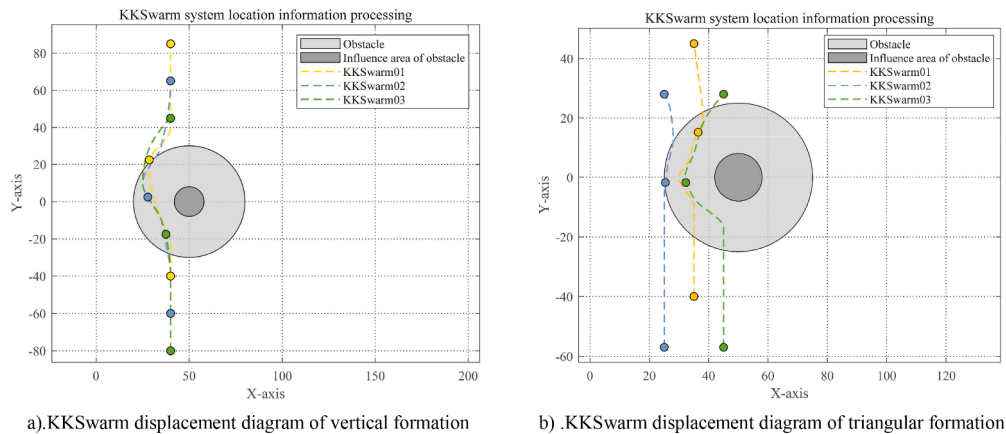


Fig. 32. KKSwarms displacement change diagram.

the AUV. The solution process is complicated, which is suitable for off-line planning underwater formation navigation task, so there are still some problems in the practical application of this algorithm. Due to limited experimental equipment, KKSwarms platform was used in this paper to verify the effectiveness of the proposed algorithm in 2D environment, while simulation verification in 3D environment was lacking. In the future, we will work on the real world AUV cluster formation simulation to better verify the formation algorithm. And we will focus on how to solve the position of underwater vehicle formation in real time, and consider the impact of signal interference, delay and packet loss caused by complex underwater environment on underwater vehicle formation.

In conclusion, the combination of virtual structure method and artificial potential field method proposed in this paper can better realize multi-AUV cluster formation obstacle avoidance, which provides a safer and more reliable formation obstacle avoidance method for underwater vehicles.

Author contributions

Concept and design: Wenjun Ding and Lei Zhang; data collection and analysis: Guozong Zhang, Chiyu Wang, Tihao Yang and Yajun Chai; drafting of the article: Wenjun Ding and Lei Zhang; critical revision of the article for important intellectual content: Wenjun Ding and Zhaoyong Mao; study supervision: Wenjun Ding and Zhaoyong Mao. All the authors approved the final article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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